

Kunyi (Mark) Ma

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Education

Columbia University, New York, NY

Ph.D. Candidate in Mathematics

2024–2029 (to be expected)

M.A. in Mathematics, 4.15/4.

2024–2026

Advisors: Daniela De Silva, Ovidiu Savin

Research interest: PDE Analysis, Free Boundaries

New York University Shanghai, Shanghai, China

B.S. in Honors Mathematics and Data Science, magna cum laude, 3.897/4

2020–2024

Existence of Stationary Measure for Incompressible 2D Stochastic Navier-Stokes Equations (with Vahagn Nersesyan)

On modifying Loss Function for PINO to approach Mathematics Nature in Fluid Mechanic (with Zhuang Su)

Nanyang Model High School, Shanghai, China

2017–2020

Talks

Online PDE Coffee Chat

Introduction to Elliptic PDEs Part II : Schauder for Uniformly Elliptic Non-divergence Equation

Mar.21st 2026

Introduction to Elliptic PDEs Part I : Dirichlet Problem for Laplace and Poisson's Equation

Mar.14th 2026

Poster Presentations

Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburg

Nov.15th 2025

Elliptic PDE Group Meeting @ Columbia

Caffarelli-Cabr : Jensen's Approximation and Uniqueness of viscosity solution to Dirichlet BVP

Feb.11th 2026

Alt-Caffarelli: Viscosity Solution, Non-degeneracy, Lebesgue Density Estimate

Nov.7th 2025

Alt-Caffarelli: Euler-Lagrange Equations, Optimal Regularity

Nov.6th 2025

The Obstacle Problem: Singular Set, Monneau Monotonicity, ACF

Oct.14th 2025

The Obstacle Problem: Classification of Blowups

Sep.30th 2025

The Obstacle Problem: Blowups, 2-homogeneity, Convexity

Sep.23nd 2025

The Obstacle Problem: Minimizers, EL equations, Continuity, Optimal Regularity, Nondegeneracy

Sep.9th 2025

Student Analysis Seminar @ Columbia: The Theory of Bounded Variation Functions & Regularity Analysis of the Minimal Surface PDE

De Giorgi's Lemma for C^1 Caccioppoli Sets

Nov.6th 2024

De Giorgi's Lemma for Harmonic and C^1 function

Oct.16th 2024

Regularity of the reduced boundary

Sep.25th 2024

Basic properties of functions of bounded variation

Sep.11th 2024

Informal Reading Seminar on GR @ Columbia

Local Theory for Quasi-linear Wave Equations

Nov.18th 2024

Existence of Solutions to non-constant coefficient linear wave IVP

Oct.7th 2024

Representation formula using Homogeneous Distribution

Sep.16th 2024

Teachings

Teaching Assistant

Applied Functional Analysis APMA4150, Columbia University

Spring 2026

Ordinary Differential Equations UN2030, Columbia University

Spring 2026

Analysis & Probability I GR6151 (Graduate Level), Columbia University

Fall 2025

Linear Algebra S2010, Columbia University

Summer 2025 B

Analysis and Optimization S2500 (Grader), Columbia University

Summer 2025 B

Introduction to Modern Analysis I S4061, Columbia University

Summer 2025 A

Directed Reading Program

Introduction to Elliptic PDEs and Geometric Measure Theory (with Liam Dennehy, Marlet Ma, Barnett Pesin) Spring 2026

Student Tutor

College Algebra and Calculus I–III, Columbia University	Fall 2024
ARC Learning Assistant for Pre-Calculus, NYU Shanghai	Fall 2021

Honors and Awards

NYU Shanghai

Magna cum laude; Phi Beta Kappa; NYU Shanghai Excellence Award; NYU Honors Scholar: Founders Day Award; 2023 NYU Shanghai Dean’s Undergraduate Research Funding; 2022 NYU Shanghai Hack the Pearl, Champion of Data Modelling Track; 2020–2022 NYU Shanghai Recognition Award; 2020–2023 NYU Shanghai Dean’s List

Attendance

SIAM Conference on Analysis of Partial Differential Equations (PD25)	Nov. 2025
Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburg	Nov. 2025
The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Québec, Canada	July 2025
PDE Summer School 2025 Theory and Applications of Elliptic PDE, UC Irvine	June 2025
Rocky Mountain Mathematics Consortium Summer School 2025, ASU, University of Wyoming	June 2025
Duke Mathematics Journal 2025 Conference, Duke University	Apr. 2025
Winter School: Boundary and Singularity in Fluid Mechanics, Simons Center	Jan. 2025
Differential Geometry Summer School 2024, PKU BICMR	July-Aug. 2024
Workshop on Minimal Surfaces and Mean Curvature Flow 2024, PKU BICMR	July 2024
International Congress of Chinese Mathematicians 2022, THU YMSC	July-Aug. 2022

Extracurricular

Orientation Ambassador, NYU Shanghai	May–Sep. 2021
President, Literature@NYUSH Student Club, NYU Shanghai	Jun. 2021–May 2022
Tenor, NYU Shanghai Elite A Cappella Chamber Singers	Feb. 2021–Dec. 2022
Quantitative Researcher, Ipsos China, Shanghai	Jun.–Aug. 2021